

PENGWIN 2026 Task 1 -- V0 Pipeline-Test Submission

Algorithm Description · 2026-05-29

0. Submission intent

This is a **V0 pipeline-test** submission for the PENGWIN 2026 Task 1 preliminary phase. The objective is to validate the end-to-end Grand Challenge contract (Docker container build, model tarball, /input MHA read, /output MHA write under `--network none`) using the current best in-house checkpoint. It is **not** a competitive scientific submission and should not be interpreted as our final leaderboard entry. The femur anatomy (label range 151-200) is **not modeled** in this V0 and is emitted as all-zero on femur-routed cases; pelvic-routed cases run the trained model.

1. Method overview

The pipeline is a single-stage nnU-Net ResEnc-L 3D full-resolution segmenter trained with an **ABBC 4-class semantic head** (background / border / boundary / core) on pelvic ROIs. Per-instance fragment labels are recovered at inference time by a **core-seed watershed** decoder operating on the softmax volume, followed by anatomy-aware connected-component pruning and a small-fragment merge step.

Components:

- **Backbone / planner:** nnU-Net ResEnc-L (nnUNetResEncUNetLPlans), config `3d_fullres`.
- **Semantic head:** 4-class ABBC softmax (background, border, boundary, core).
- **Instance decoder:** `task1_v288_abbc_core_seed_watershed` (V288 decoder reused from internal eval CLI).
- **Anatomy routing:** spec-provided rule (see Section 3) selects pelvic vs. femur; V0 only handles pelvic.

2. Training

Item	Value
Dataset	Dataset537_PelvicBICMFragmentV5 (Sacrum / LeftHip / RightHip; no Femur)
Trainer	PengwinTrainerBoundaryFragmentSidecarCoreRecallDenseCandidateCore025StrongPeakNoContactABBCV291
Plans	nnUNetResEncUNetLPlans
Config	3d_fullres
Fold	fold_0 only
Boundary class weight (bw)	10
Epochs	50
Iterations / epoch	100 train / 20 val
Train ROIs	12 stratified (sub-fulltrain probe)
Val ROIs	6 stratified
Best EMA val score	0.188 (best checkpoint at epoch 34 spike)
Checkpoint	checkpoint_best.pth (~819 MB)

The model was trained as a **probe** for the V291 ABBC head design, not as a full-data training run. A Phase 6 fulltrain on all 174 cases is planned but is not part of this submission.

3. Inference

3.1 Anatomy routing (verbatim from spec)

```
def classify_pelvic_femur(spacing_x, spacing_y, spacing_z,
                        physical_x_mm, physical_z_mm):
    if physical_x_mm <= 285.35:
        if spacing_x <= 0.71:
            return "pelvic"
        elif spacing_z <= 0.90:
            return "femur"
        else:
            return "pelvic" if spacing_y <= 0.91 else "femur"
    else:
        if spacing_z <= 0.68:
            return "pelvic" if physical_z_mm <= 193.55 else "femur"
        else:
            return "pelvic" if physical_z_mm <= 390.78 else "femur"
```

For V0: pelvic cases run the trained model; femur cases are short-circuited and an all-zero label map matching the input spacing / dimensions is written to /output.

3.2 Pelvic pipeline

1. **Read** input MHA from /input/images/peripelvic-fracture-ct/.
2. **Anatomy classification** via the rule above.
3. **nnU-Net ResEnc-L predict** with the V291 bw=10 fold_0 checkpoint, producing the ABBC 4-class softmax volume.
4. **Core-seed watershed** (V288 decoder) lifts the softmax volume to per-instance integer fragments.
5. **Connected-component prune** at $\sim 1 \text{ cm}^3$ to remove speckle.
6. **Anatomy-size-ratio merge** with --anatomy-size-ratio-keep 0.05 (small fragments below 5 % of the largest fragment within an anatomy are merged into the nearest large fragment).
7. **Label remap** to the PENGWIN 2026 official ranges: Sacrum 1-50, Left Hipbone 51-100, Right Hipbone 101-150, Femur 151-200 (femur is unused in V0).
8. **Write** integer 3D label map MHA to /output/images/peripelvic-fracture-ct-segmentation/, preserving input spacing and dimensions.

The runtime entrypoint is inference/inference.py, packaged at /opt/app/ in the container. The checkpoint, plans.json, and dataset_fingerprint.json are shipped via the model tarball, extracted by Grand Challenge under /opt/ml/model/.

4. Known limitations

- **Femur not modeled.** Femur-routed cases output an all-zero label map. Any case whose ground-truth contains labels in [151, 200] will score zero on those instances.
- **Sub-fulltrain probe.** Only 12 of 174 cases were used for training; the EMA val score (0.188) and the hard-trio metrics (below) are well below the leaderboard target. Hard-trio Fracture Dice 0.812 is below the ~ 0.90 leaderboard expectation.
- **Single fold, single seed.** No ensembling, no TTA beyond nnU-Net defaults.
- **Phase 6 fulltrain planned.** The next submission will train on all 174 cases with both pelvic and femur anatomies and a full 1000-epoch schedule.

5. Hard-trio Phase 5.5 v2 metrics

Metrics are computed by our internal CLI task1-v288-eval-v2 (compute_task1_official_aligned_v2_metrics), which is aligned to the PENGWIN 2026 official scoring spec. The hard-trio is cases 003, 011, 017.

The headline V0 configuration is the last row (**bw=10, anatomy-size-ratio-keep = 0.05**):

Config (bw / sr_keep)	Fracture Dice	Local Dice	Instance F1	HD95 (mm)	ASSD (mm)	Merge Err	Split Err	Topology Cons.
bw 2.5 / sr 0.00	--	--	--	--	--	--	--	--
bw 2.5 / sr 0.05	--	--	--	--	--	--	--	--
bw 10 / sr 0.00	--	--	--	--	--	--	--	--
bw 10 / sr 0.05 (V0)	0.812	0.819	0.890	11.79	2.59	2	21	0.444

Cells marked -- are alternate ablation configurations that were run internally but are not reported in this V0 submission; only the bw=10 / sr=0.05 row is the configuration actually shipped in the container.

6. Pointers

- Anatomy routing rule: copied verbatim from the PENGWIN 2026 Task 1 specification.
- Reference baseline: YzzLiu/PENGWIN2026_Task1_AutoSeg_Baseline (preprocessing / training / inference scaffolding; no evaluate.py).
- Internal eval CLI used to produce the metrics in Section 5: task1-v288-eval-v2 (compute_task1_official_aligned_v2_metrics).
- Internal predict CLI mirrored by the container entrypoint: task1-v288-predict (code_task1/eval.py, --cases / --out-root / --checkpoint / --fold / --trainer / --plans / --config / --preprocessed-plans-dir).